Exploring the current and future Scenario of Port Blair: A Simulation-Based Analysis of Harbour Operations

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Executive summary:

Discrete Event Simulation model was used to analyse the Port Blair's harbour's capacity under both the Business-as-Usual and Future Scenarios. The answers to the key questions of interest are as follows:

1. **Is the harbour's capacity sufficient to handle current demand**
   Yes, the harbour size is sufficient to handle the current capacity.

2. **Would upgrading the lock or unloading processes within a 1.5x improvement limit enhance efficiency under current conditions?**
   Under Business-as-usual scenario, the system harbour is never reaching its maximum capacity and the number of ships waiting outside the lock never grows. Hence, the improvements are not necessary.

3. **Under Future Scenario if arrivals of the ships are expected to increase by 10 percent, then, what happens to the harbour without any improvements? Are improvements necessary and will they help handle the capacity?**
   Under future scenario if the rate of arrivals is increased by 10 percent, the occupancy in the system increases slightly but the system [1896438] is smooth enough to handle this rate of increase. Hence, improvements are not necessary.

Introduction:

The system under study is Port Blair's harbour, which can dock one ship at a time. It has a harbour capacity of four where, which allows four ships to wait any time. There is a lock between the harbour and sea that allows the ship to enter the harbour when there are less than four ships in the harbour. If the ships in the harbour are more than four then the lock does not allow the ship to enter the harbour, the ship remains in the lock. At a given time, only one ship can be in the lock, any other ships wait outside in the sea if there is a ship in the lock. These ships wait outside the lock until the lock becomes free.

The primary aim of this study is to answer the following questions of interest under two scenarios business as usual (BAU) and Future scenario (FS).

1. Is the harbour's capacity sufficient to handle current demand?
2. Would upgrading the lock or unloading processes within a 1.5x improvement limit enhance efficiency under current conditions?
3. Under Future Scenario if arrivals of the ships are expected to increase by 10 percent, then, what happens to the harbour without any improvements? Are improvements necessary and will they help handle the capacity?

To answer the questions of interest, a prebuilt Discrete event simulation model was used, and verification, validation, accreditation & acceptance was done to ensure the simulation model is functioning as per the requirements. To perform VV&A, a schematic diagram, flowchart, event table and state transition diagram were developed based on the physical structure of the system. Certain parts of the code were changed based on VV& A and historical data to ensure that the system if functioning as per the requirements and is reliable to get the answer to the questions of interest.

Before addressing the questions, the burn-in time was determined using Welch's method, and this period was removed from the observations to answer the questions. Time average and ensemble average of the sea and harbour queue was used to answer the first two questions. To answer the third question ensemble average was plotted for the whole system and from this claim is made that there is no improvement required even if the rate of arrivals increase by 10 percent.

Background

Discrete Event Simulation is a technique used to model real-world systems by simulating their behaviour through events that occur over discrete time. It provides a detailed analysis of how a system operates and performs, offering valuable insights for decision-making. Over time, DES has become an essential tool in various industries due to advancements in computing technology. Discrete Event Simulations are extensively used to solve a wide range of problems in the real world. They are used to model hospital arrivals to ensure the ICU capacity, and opening of a new restaurant to ensure the amount of space required for parking and seating, etc.

In this simulation two FIFO queues were used which allow the first arriving entity to get served before all the arriving entities arriving after it.

Method:

Analysing Historical Data:

The port authorities provided historical data which contained ship interarrival, locking and unloading times. This historical data was cleaned[1], and histograms were plotted to select the distribution family. Method of matching moments was used to estimate the parameter values. After obtaining the parameter values, QQ-plots were generated to determine which distribution matches the historical data. The distributions with the parameters who's theoretical and sample quartiles formed nearly a straight line were selected for generating process times.

Interarrival times

Figure 4 shows the histogram of interarrival times generated using historical data and to verify the type of distribution, uniform and normal distribution types were used. Moments were calculated for interarrival times and the values of first and second moment are 36.167 and 1429.957[2]. Based on these values, the lower and upper boundary values for interarrival times were calculated as 11 and 59 respectively[3]. For generating normal distribution, the values of mean and standard deviation were evaluated as 35.167 and 13.901[4] these values were rounded as 35 and 14 to generate QQ-plot. Figure 5 shows QQ-plots.

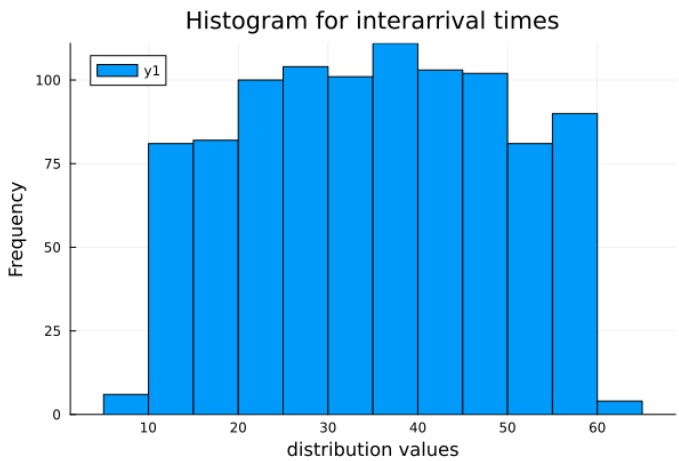

Figure. 4 Histogram of interarrival times from historical data[5]

[1] Appendix point 1
[2] These values are in output of Terminal_output.txt file present in terminal data on line 8 and 9, generated by calling moments function present in the functions file on line 318, it is called on line 63 in harness file.
[3] See appendix point 2 for calculation of upper and lower boundaries for uniform distribution.
[4] These values are in output of Terminal_output.txt file present in terminal data on line 12. These values are calculated using the command on line 69. Standard deviation is not rounded in the terminal output file to three decimals.
[5] The plot generated by code in harness file on line 55 and is saved in figures/interarrival times/Histogram.png.

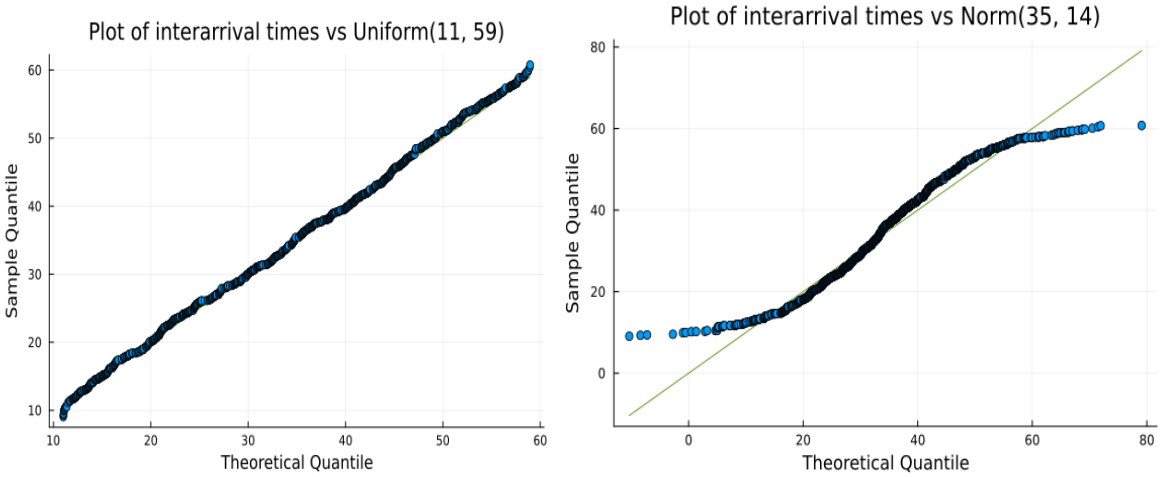

Figure. 5 QQ-plots for interarrival times against Uniform (10, 60) and Normal (35, 14) [6]

Lock times:

The histogram for lock times is shown in Figure 6[7]. The distribution shape suggests that the data likely belongs to either the exponential or Pareto distribution. Moments were calculated for lock times from the distribution as 10.928 and 236.321[8]. Exponential distribution with mean 11 was used, as the first moment is the mean, and for pareto distribution, α and β values were calculated as 2 and 6[9] and were used to plot two QQ-plots to determine the distribution for lock times. The QQ-plots for lock times are shown in figure 7[10].

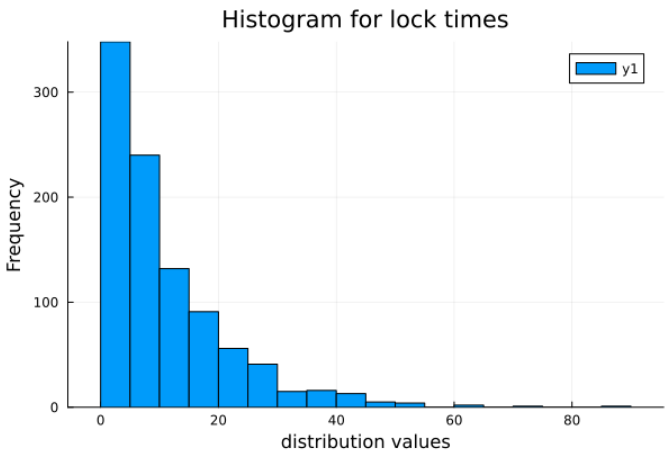

Figure. 6 Histogram of lock times from historical data

[6] Plots are in figures/interarrival times as VsNorm(35, 14).png and VsUniform(11, 59).png, generated using commands on lines 76 and 83 in harness file.
[7] This plot is in figures/lock times/Histogram.png and generated using command on line 97 in harness file.
[8] These values are in output of Terminal_output.txt file present in terminal data on line17, generated by calling moments function present in the functions file on line 318, it is called on line 105 in harness file.
[9] Calculation for alpha and beta is shown in Appendix point 3.
[10] Plots are in figures/lock times as VsExp(11).png and VsPareto(2, 6).png, generated using commands on lines 113 and 120 in harness file.

From these plots, it was decided that the locking times data belongs to exponential distribution as the QQ-plot for pareto distribution is not a straight line while that for exponential distribution is a straight line.

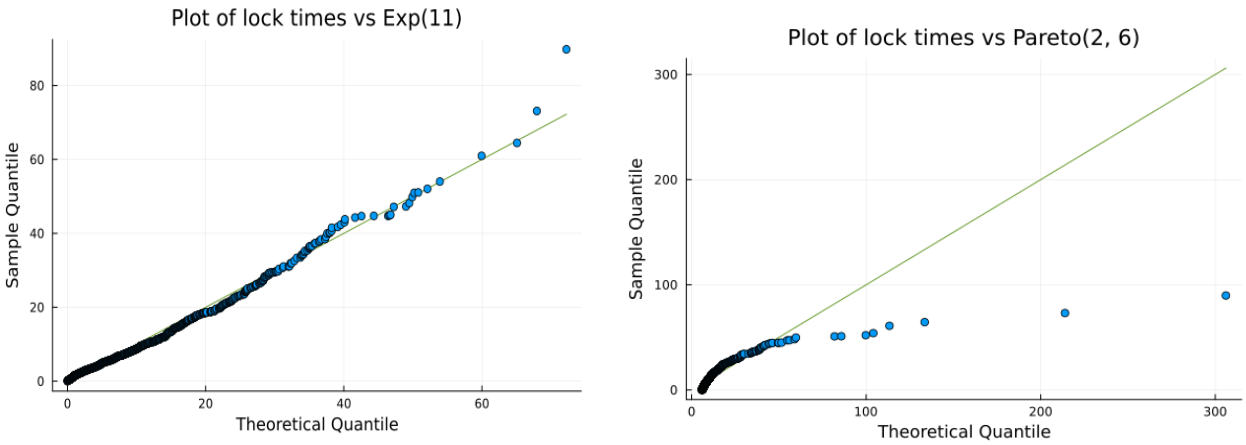

Figure. 7 QQ-plot for lock times against Exponential (11.0) and Pareto (2, 6)

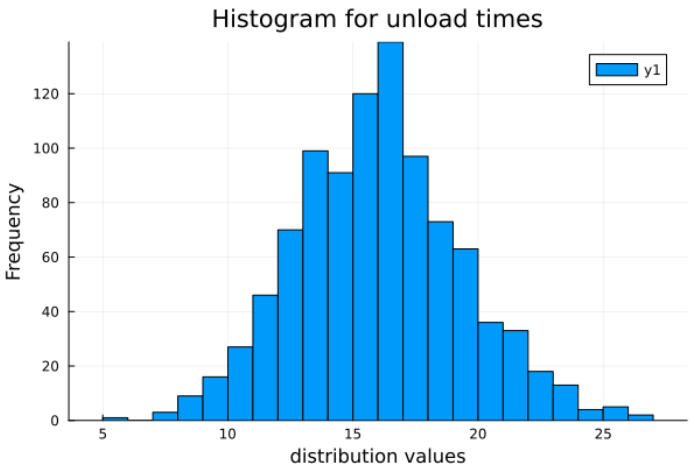

Figure. 8 Histogram of unload times from historical data

Unload times

Unloading times are normally distributed as shown in figure 8[11]. The values of first and second moments were calculated for unloading times and they are 16.021 and 267.572[12] Using these moments, the mean and standard deviation were obtained as 16.021 and 3.301[13],

[11] Plot is in figures/unload times as Histogram.png and is generated using commands on lines 134 and 135 in harness file.
[12] These values are in output of Terminal_output.txt file present in terminal data on line 22, generated by calling moments function present in the functions file on line 318, it is called on line 142 in harness file.
[13] For mean and standard deviation output see line 23 in Terminal_output.txt file located in terminal data folder and command to calculate this is in harness file on line 154.

these values were rounded to 16 and 3 respectively to generate normal distribution. No additional distribution families were considered since the QQ-plot of the unload times from the historical data formed a straight line as shown in figure 9[14]. This pattern, along with the observed shape of the distribution, provides sufficient evidence to conclude that the unload times follow a normal distribution

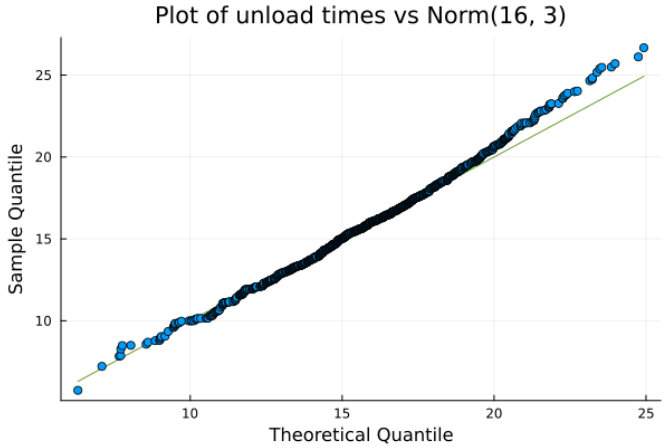

Figure. 9 QQ-plot for unload times against Normal (16, 3)

Schematic diagram:

Schematic diagram is shown where sea queue is represented as a FIFO queue. This queue has infinite capacity. If the lock is busy, then the ship from the sea queue cannot enter the lock as there is only one space, ships can only enter the lock when the lock is free. Also, this lock only allows the ship that is inside the lock to enter the harbour queue, when the number of ships in the harbour queue is less than four. Harbour queue size is restricted to four, so at any time interval at most four ships are inside the harbour queue. Harbour queue also follows FIFO mechanism. After the harbour queue, there is docking site where the docking of a ship is done. Only one ship can be docked at any time interval. After the docking is completed, the ship is removed from the system and next waiting ship from the harbour is allowed to enter the docking site. In this system there are two queues in series, with lock and docking site as two servers.

[14] Plot is in figures/unload times as VsNorm(16, 3).png, and is generated using commands on lines 148 to 150 in harness file.

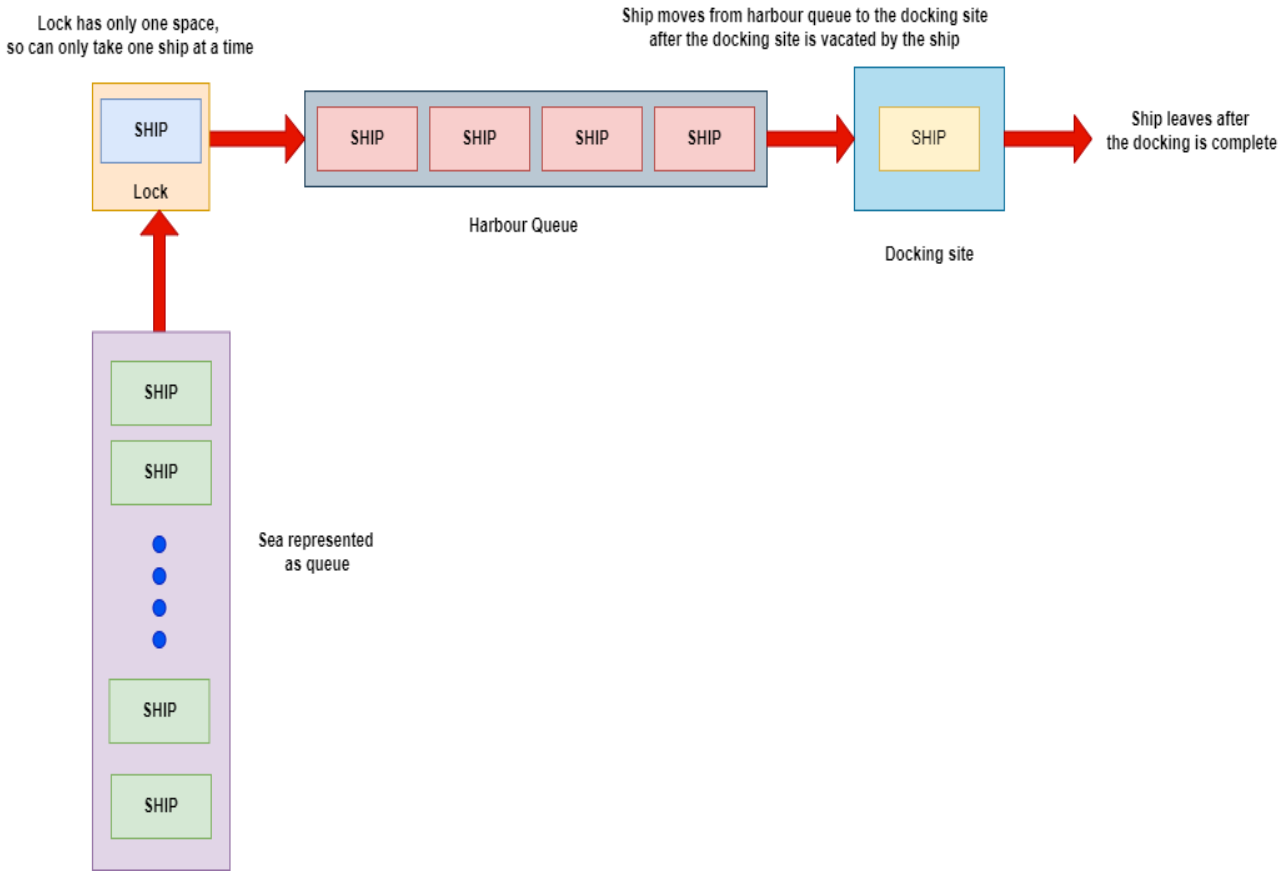

Figure 1. Schematic diagram of the system

Flowchart:

Flowchart of the simulation is shown in figure 2. The system is initialized with first arrival, then the simulation is run for time, which is checked before any event is dequeued from the event queue. If the time exceeds the time horizon, the simulation is terminated else, next event is dequeued from the event queue and it is processed.

Arrival Event

If the event is an arrival event, then next arrival is added to the event queue and this arrival ship is added to the queue which is before the lock also known as sea queue. After this the availability of lock is checked, if available, then the next waiting ship in the sea queue is dequeued and added to the lock server, a departure event is scheduled for this ship and this event is added to the event queue. If the lock is busy, no action is performed.

Departure Event

If the event dequeued from the event queue is departure event, then check is performed to know at which server, the departure is performed. If it is lock, then the algorithm checks that, the size of the harbour queue is less than four. If the size is less than four, then the ship is removed from the lock and added to the harbour queue and a new ship is added to the lock

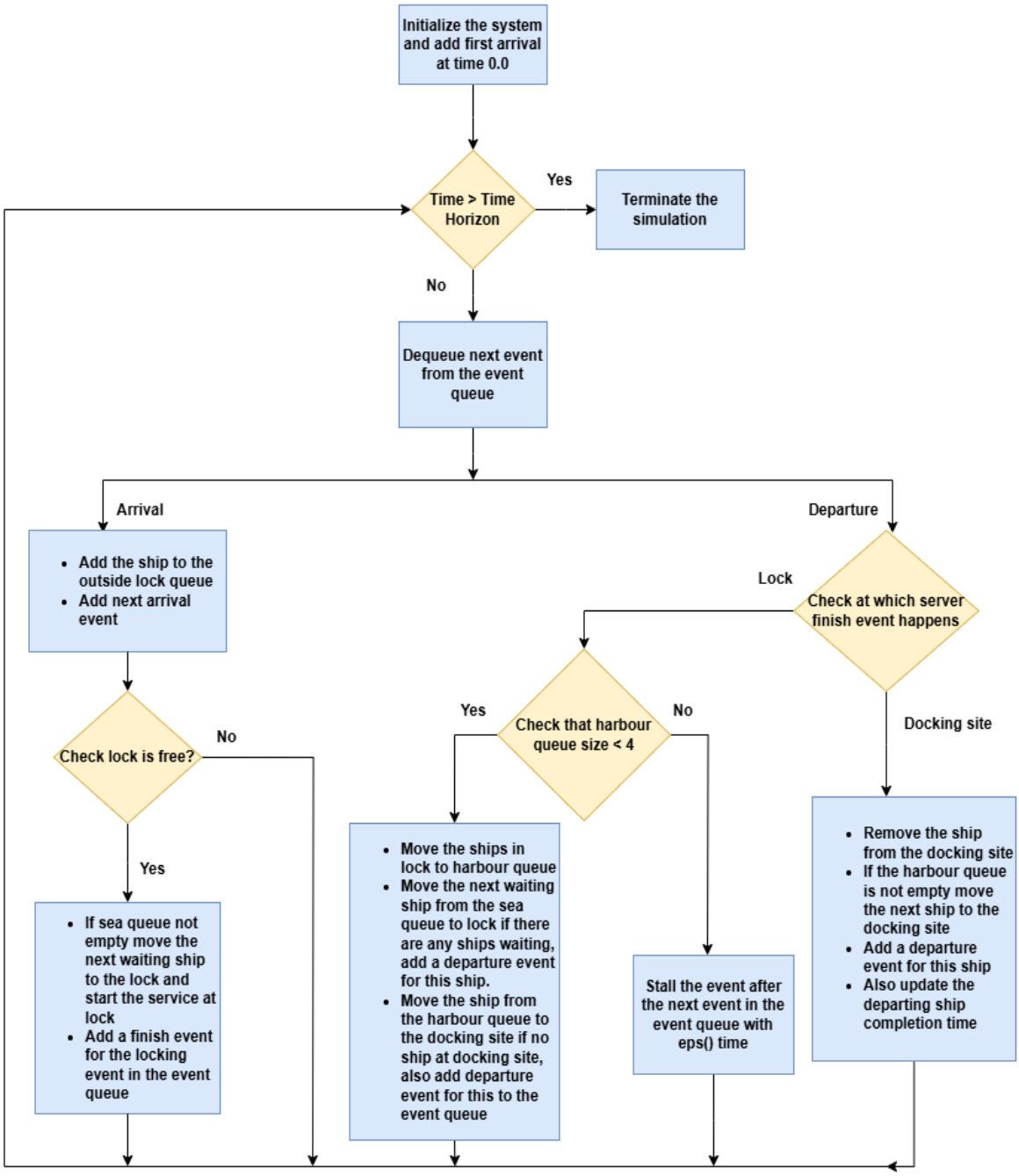

Figure 2 Flowchart of the harbour simulation

for locking from the sea queue if there is any ship waiting in the sea queue. If there are already four ships waiting in the harbour queue for docking, and a departure event is scheduled, then the departure event is stalled with epsilon time interval and the ship remains in the lock. If the departure event happens at the docking site, the ship is removed from the docking site, and the completion time is updated, also if there are ships waiting in the harbour queue, then they are added to the docking site for service.

State Transition Diagram:

Figure 3 shows how state transition diagram of the system.

- When there is one ship in lock and zero or one ship in dock, and zero ships in harbour queue, and an arrival event is triggered, then the sea queue begins to grow, this is shown with blue arrows in the figure 3 when there are zero ships in the harbour, the sea queue grows. Another possibility is when, the lock has one ship, dock has one ship and harbour queue can have 1, 2, 3, or 4 ships, and an arrival happens then, also sea queue grows. This is also shown as blue line in figure 3 when ships in harbour queue are between 1 to 4.
- The green arrows show that lock has one ship, dock site has zero ship, and a departure event is triggered at the lock, then the ship from the lock is moved to the dock and next waiting ship from the sea queue is added to the lock for service.
- Yellow arrows show that when there is one ship in the lock, and one ship in the dock site and departure at lock is triggered. This transition can only happen when the number of ships in harbour is less than 4.
- Magenta coloured arrows show state transition when,
    1. Zeros ships in the sea queue, one ship in dock and one or zero ship in the lock, and a departure at the dock is triggered, then the ship is removed from the dock and next ship is added to the dock from the harbour queue.
    2. More than or equal to one ship in the sea queue, one ship in both lock and dock, and a departure at the dock server happens, again the ship is removed from dock and next waiting ship from harbour queue is added.

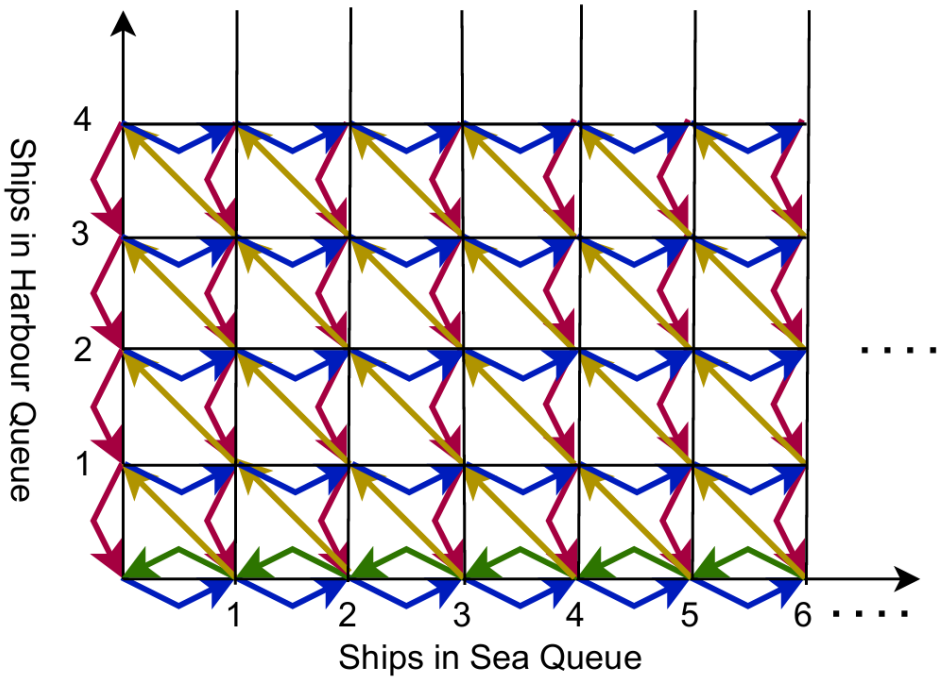

Figure 3 State transition diagram of the harbour simulation

Event Table:

Time	Event, event id	(Sea queue, Harbour Queue) before	(Sea queue, Harbour Queue) after	Lock (before, after)	Dock (before, after)	Event Queue After an event occurs (Event, time, event id)
0.000	Arrival, 0	(0, 0)	(0, 0)	(0, 1)	(0, 0)	(Finish (1), 7.425, 2), (Arrival, 39.089, 1)
7.425	Finish (1), 2	(0, 0)	(0, 0)	(1, 0)	(0, 1)	(Finish (2), 23.720, 3), (Arrival, 39.089, 1)
23.720	Finish (2), 3	(0, 0)	(0, 0)	(0, 0)	(1, 0)	(Arrival, 39.089, 1)
39.089	Arrival, 1	(0, 0)	(0, 0)	(0, 1)	(0, 0)	(Finish (1), 41.152, 5), (Arrival, 81.487, 4)
41.152	Finish (1), 5	(0, 0)	(0, 0)	(1, 0)	(0, 1)	(Finish (2), 54.611, 6), (Arrival, 84.487, 4)
54.611	Finish (2), 3	(0, 0)	(0, 0)	(0, 0)	(1, 0)	(Arrival, 84.487, 4)

84.487	Arrival, 4	(0, 0)	(0, 0)	(0, 1)	(0, 0)	(Finish (1), 96.503, 8), (Arrival, 94.502, 7)
94.502	Arrival, 7	(0, 1)	(0, 0)	(0, 1)	(0, 0)	(Finish (1), 96.503, 8), (Arrival, 126.783, 9)
96.503	Finish (1), 8	(0, 1)	(0, 0)	(1, 0)	(0, 1)	(Finish (2), 119.782, 10), (Arrival, 126.783, 9),

Table. 1 Event table for harbour simulation.

Assumptions

- Interarrival times are uniformly distributed with lower and upper values as 11 and 59 respectively.
- Lock times are exponentially distributed with mean 11.
- Unloading times follow exponential distribution with mean 16 and standard deviation 3
- All process times are independent of each other.
- Harbour and sea queue both follow FIFO mechanism.
- Size of harbour queue is restricted to 4, so at any time there can be at most 4 ships in the harbour queue.
- Size of sea queue is infinity, and the ships wait until they arrive at the lock for service.
- There is no pre-emption allowed and there is no priority for ships all ships are considered same.
- Lock does not allow ships to enter harbour queue, if there are four ships waiting in the harbour queue to get served.
- Lock and dock both can process only one ship at a time.
- Departure at the dock removes the ship from the system.

Burn-in period

Using Welch's method the burn-in time was selected by first calculating windowed average because Welch's method requires same number of observations (Mahajan, P & Ingalls, R 2004). After this ensemble average was calculated over 100[15] realizations. Using this

[15] This operation is on line 471 in harness file uses seeds from 1896438 to 1896537 and ensemble average is calculated inside burnin_time() function in the fns file on line 347.

ensemble average, moving average was calculated and plotted on the graph as shown in figure 10[16]. From this, the value of burn-in was selected as 3000 because after this value the moving average converges.

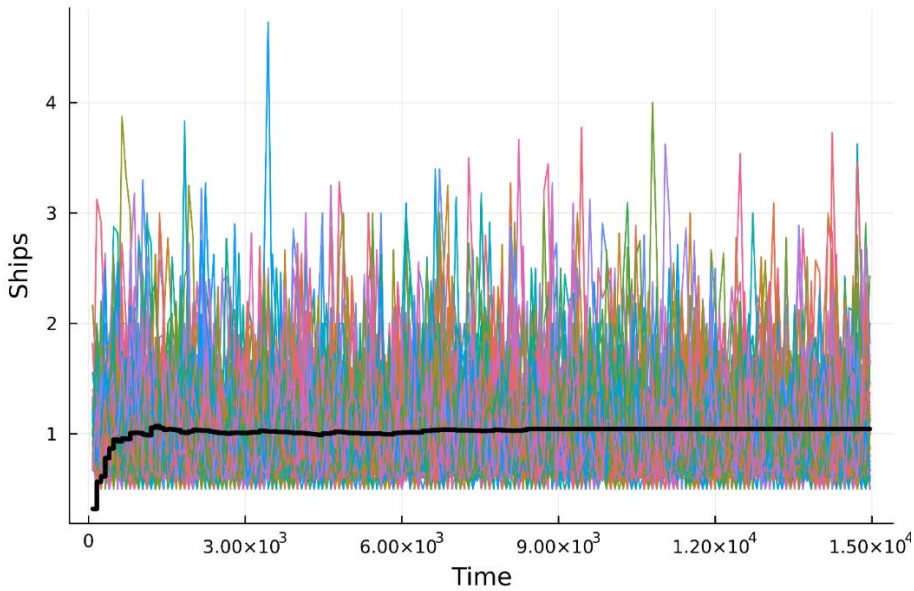

Figure 10 Estimating burn-in time

Selection of number of realizations and Time horizon

200 realizations are used to answer the questions as the amount of time used to generate one realization was approximately 1.2749 seconds and so 200 realizations will take approximately 254.98 seconds and increasing the number of realizations increased the computational load. Once the burn-in time was determined, the time horizon was set to 30,000 hours. This duration was chosen to ensure enough data for analysis.

[16] Plot is in figures/burnin times as burnin_time.png and is generated using command on line 471 in harness file, which calls a function on line 347 in the functions file to create the plot.

Results:

Question 1

The histogram in figure 11 is time average computed and this was done for 200 realizations[17]. The burn-in time was excluded and the observations between 3000 to 30000 were used from the state data file to calculate the time average number of ships in the harbour queue. The time average was calculated as 0.0916[18], and the 99 percent confidence intervals were calculated as 0.0687 and 0.1234[19]. From this histogram it is clearly visible that the harbour hardly accommodates any ship in the harbour queue the greatest number of times. So, the size of harbour is large enough. Also, ensemble average (black line) was plotted with 24 hours' time frame and the plot is as shown in figure 12[20]. The ensemble average is very less, and it is less than one hence the shape also favours the claim that the size of harbour is large enough for the ships to wait to be unloaded.

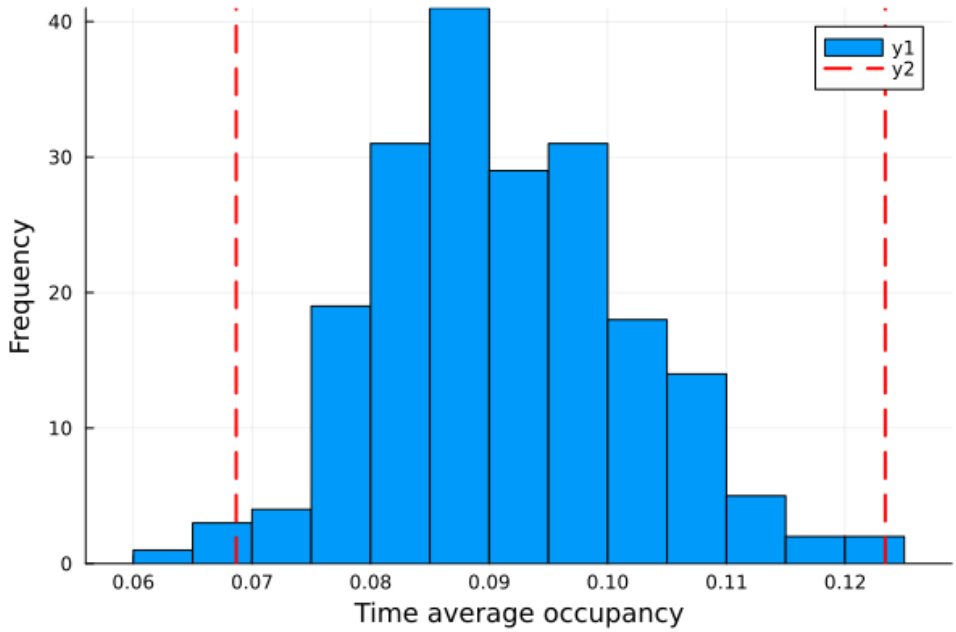

Figure 11 Time average occupancy of harbour queue for 200 realizations.

[17] Harbour harness file 485 lines show that 200 seeds are used, and histogram is generated for this time average. Plot in QUESTIONS/Question1A.png. histogram command on line 493 harness.

[18] These values are in output of Terminal_output.txt file present in terminal data on line 47, harness file 502

[19] These values are in output of Terminal_output.txt file present in terminal data on line 48, harness file 495 and 503, cf() called from run file on line 436.

[20] Plot in QUESTIONS/Question1B.png and generated using commands on line 505 in harness file and uses q1() in fns file on line 442.

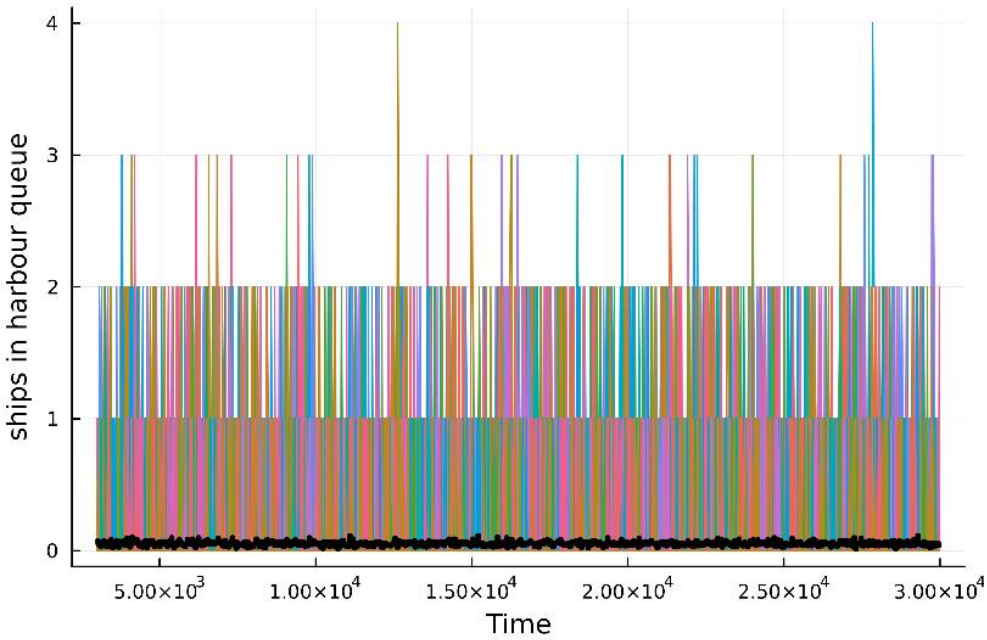

Figure 12 Ensemble average plot of harbour occupancy every 24 hours plotted using 200 realisations.

Question 2

From the analysis done in previous question, it is concluded that there is no need to upgrade the dock as the size of harbour queue does not grows large. To further investigate that the upgrade is required at lock, time average of sea queue after every event was calculated for 200 realizations and the values are plotted as histogram in figure 13[21]. Time average occupancy of the sea queue for 200 realizations over 27000 hours was calculated as 0.045[22] with 99 percent confidence intervals as 0.026 and 0.078[23] respectively. From, the above results the sea queue also remains vacant, and the time average waiting numbers are very less. Also, ensemble average was plotted with 24 hours' time frame, and the plot is as shown in figure 14[24]. The ensemble average (black line) is very less, and it is less than one hence from the plot of ensemble average, we claim that for Business as usual there is no need to upgrade the lock also. The system does not need any improvements as the sea queue and harbour queue hardly grows.

[21] Plot in QUESTIONS/Question2A.png and generated using commands on line 520 in harness file.
[22] Line 51 Terminal_output.txt file, command on harness file line number 529.
[23] Line 52 Terminal_output.txt file, command on harness file line number 530, cf() called from run file on line 436.
[24] Plot in QUESTIONS/Question2B.png and generated using commands on line 532 in harness file, q2() in fns file line 469.

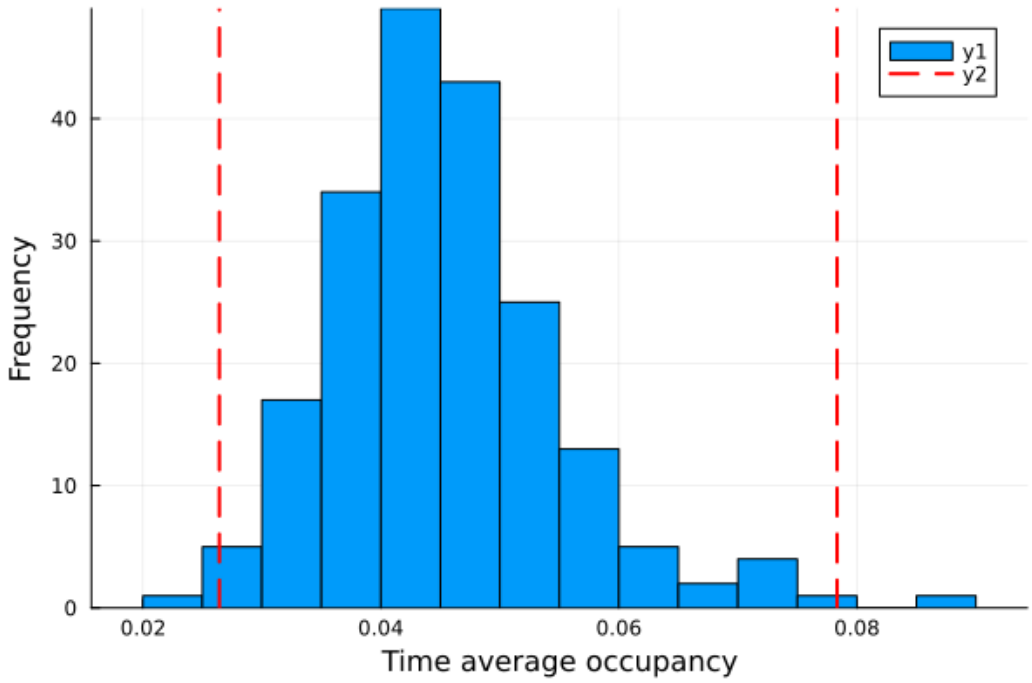

Figure 13 Time average occupancy of sea queue for 200 realizations

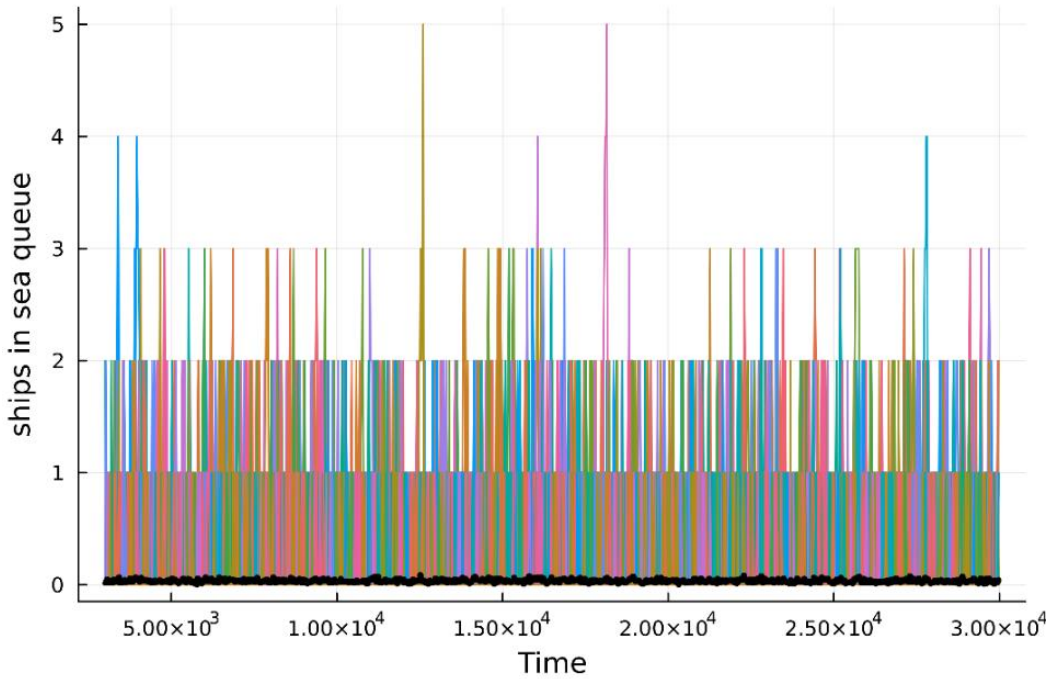

Figure 14 Ensemble average plot of sea occupancy every 24 hours plotted using 200 realisations.

Question 3

Figure 15[25] shows ensemble average with 24 hours' time frame, this average is calculated by addition of the total ships in harbour, sea, lock and dock. From the figure it can be visualised that when the rate of arrivals is increased by 10 percent, there is no significant changes in the occupancy of the system from previous scenario (BAU). The system still accommodates ships in the range 0.8 to 1.2[26] for the future scenario, which has slightly increased from Business as usual. Hence, from this we claim that event if the arrivals are increased by 10 percent the ensemble average of total ships in the system is very small and less than 2[27], which is small as compared to the total system capacity. Hence, the improvements are not necessary for future scenario.

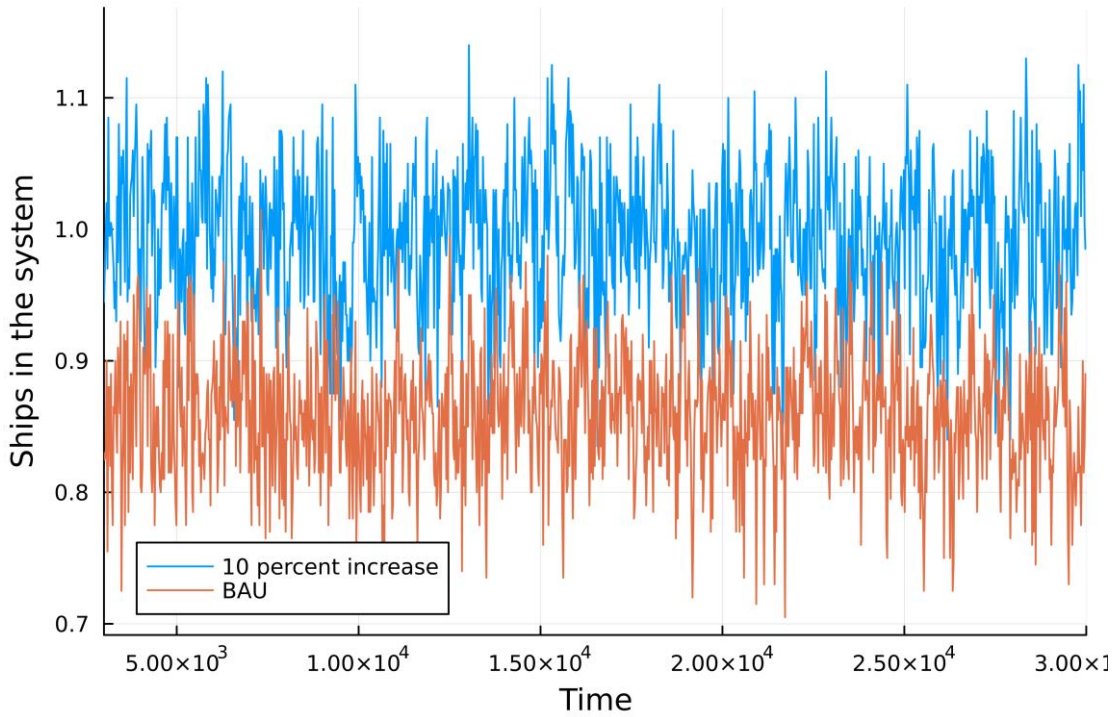

Figure 15 Plot of mean ships in the system

[25] Plot in QUESTIONS/Question3A.png and generated using commands on line 566 in harness file
[26] This is claimed using the figure 15 approximate ranges.
[27] This is claimed using the figure 15 ranges.

Conclusion

The answers to the questions of interest are as follows:

i. Yes, based on time average value and ensemble average shape, the harbour hardly accommodates one ship at any time, so from this we can say the harbour is large enough.
ii. The harbour and sea queue never grows; hence the improvements are not necessary.
iii. Even if the ship arrivals are expected to increase by 10 percent, the average number of ships in the system remain below 2. Hence, for the future scenario, the improvements are not required.

Appendix:

1. Cleaning the data
   - In terminal data there is Terminal_output.txt file where the line 3 shows the number of zeros in the historical data. The operation for this is in harness file on line 38. Instructions to run the code are given in appendix point 5
   - The cleaning is shown in line 44 of the harness file

2. m1 = 35.167 and m2 = 1429.957, (values used are rounded)

   For Uniform distribution,
   $mean = m1 = \frac{a+b}{2}$ and $m2 = \frac{a^2+ab+b^2}{3}$ (Mays, A 2024)

   $\therefore \frac{a+b}{2} = 35.167$ … (i)

   $\therefore \frac{a^2+ab+b^2}{3} = 1429.957$…(ii)

   $\therefore b = 70.334 - a$ …. From (i)

   Substituting this value in equation one and simplification gives the following equation,

   $(70.334)^2 - 70.334\,a + a^2 = 4289.871$

Solving this gives a = 11.0897 and b = 59.244. To generate a uniform distribution, these values were rounded to nearest integer. Therefore, the value of a = 11 and b = 59.

3. m1 = 10.928 and m2 = 236.32 (using 2 decimals instead of 3 for m2)

   For Pareto distribution values of α and β in terms of first and second moments are given as (Mays, A 2024),

$$\alpha = 1 + \sqrt{\frac{m2}{m2 - m1^2}} \quad \text{.... (i)}$$

$$\beta = \frac{m2}{m1}\left(1 - \sqrt{\frac{m2 - m1^2}{m2}}\right) \quad \text{.... (ii)}$$

   Substituting the values of moments,

$$\therefore\ \alpha = 1 + \sqrt{\frac{236.32}{(236.32) - (10.928)^2}}$$

$$\therefore\ \beta = \frac{236.32}{10.928}\left(1 - \sqrt{\frac{236.32 - (10.928)^2}{236.32}}\right)$$

   Solving this using wolfram alpha, the values of α and β , are 2.42 and 6.416, so the rounded values used are 2 and 6.

4. Initially the interarrival times were from uniform distribution with upper and lower values 11 and 59 respectively. Later, increase in arrival rates is estimated as follows. Arrival rates are given by number of arrivals per unit time. So, increase in arrival rate will make the mean interarrival times to reduce by 1.1. Hence, the mean interarrival which is given by equation

$$\therefore\ \frac{a+b}{2} = \frac{11+59}{2} = 35$$

   Now, dividing by 1.1 to reduce the time, so the new mean time is
   Mean interarrival (10% increase) = $\frac{35}{1.1}$ = 31.8
   So, new lower and upper limit are given as follows:

$$\therefore\ a = 31.8 - \frac{48}{2} \text{ and b } = 31.8 + \frac{48}{2}$$

Here the values are added and subtracted by 24 because the spread is not changing because rate change causes only mean to change.

$\therefore\ a = 7.8\ and\ b = 55.8$

5. How to run the code instructions:
    - Use harness file, run all the commands from 1 to 12, then run 14 to 16, then run 19 to 48, then run 55 to 85, then run 97 to 122, then run 134 to 156, the run 166 to 338, the run 346 to 399 collectively to generate data without 10 percent increase, then run 403 to 456 collectively to generate data with 10 percent increase, then run 466 to 472 collectively, then run 481 to 506 collectively, then run 511 to 533, then run 537 to 572.

    After completing this the terminal file will be written with all the necessary statements. There were some conflicting messages that is why we are running the code chunk wise to avoid those messages pop up on university computers.

6. Changes made to the functions file code:
    - Changed the distribution type for locking times from pareto to exponential on line 118.
    - Changed the distribution type for unloading type from exponential to normal on line 119.
    - Changed mean_construction_times to mean_process_times in the parameters struct on line 51 to improve readability
    - Commented the operation system.n_events += 1 in the initialise function on line 133 and added system.n_events += 1 on line 220 to ensure the events are updated with correct event id.

7. Testing the code:
    - These tests were performed and are in the harness file on lines 166 to 338, the outputs could not be printed to the terminal file for some reason, but they are running and print the output on terminal.

    i. Unit test:
    - random number generator, ship, system state, and initialise were unit tested

    ii. Integration testing:
    - Integration testing was performed to ensure that the move to server is functioning properly. Initialise, and ship were also used for this integration testing. Move to server was checked that it is moving the ship to lock and then the values were equated to ensure the system is working properly. Similarly, it was also checked that it is functioning as required when the move to server is called at the dock.

iii. **Performance testing:**

- Performance testing was also performed to look at how much time it is required to generate one realization. The time it took to run one realization was 1.2749 seconds.

iv. **Regression testing:**

- Regression testing was performed many times. This was done to ensure the functions are performing properly and the bug is fixed.

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References

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